

Question	Answer	Marks
1(a)	rate of change of velocity	B1
1(b)(i)	$s = ut + \frac{1}{2}at^2$ and $u = 0$ or $s = \frac{1}{2}at^2$ $t = \sqrt{(2 \times 14 / 9.81)}$	C1
	$t = 1.7 \text{ s}$	A1
	OR $v = \sqrt{(0^2 + 2 \times 9.81 \times 14)}$ $v = 17 \text{ (m s}^{-1}\text{)}$ $17 = (0 + 9.81) \times t$ or $17 = 9.81 \times t$ or $14 = \frac{1}{2} \times (0 + 17) \times t$ or $14 = \frac{1}{2} \times 17 \times t$	(C1)
	$t = 1.7 \text{ s}$	(A1)
1(b)(ii)	$s = ut + \frac{1}{2}at^2$ $u = (3.6 - \frac{1}{2} \times 9.81 \times 1.7^2) / 1.7$	C1
	$u = (-) 6.2 \text{ m s}^{-1}$	A1
	OR $v = (3.6 + \frac{1}{2} \times 9.81 \times 1.7^2) / 1.7$ $v = 10 \text{ (m s}^{-1}\text{)}$ $10 = u + 9.81 \times 1.7$ or $10^2 = u^2 + (2 \times 9.81 \times 3.6)$ or $3.6 = \frac{1}{2} \times (u + 10) \times 1.7$	(C1)
	$u = (-) 6.2 \text{ m s}^{-1}$	(A1)

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1(c)(i)	Less time (as it reaches a lower height)	B1
	(because) the initial <u>vertical</u> (component of the) velocity is smaller (than in part (b))	B1
1(c)(ii)	The (total) initial energy is the same (as in part (b))	B1
	change in gravitational potential energy is same, so speed is the same	B1

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